

[ANONYMIZED FOR REVIEW]

GENERATIVE AI ENHANCES INDIVIDUAL CREATIVE WRITING QUALITY BUT REDUCES THE COLLECTIVE DIVERSITY OF NOVEL CONTENT: AN EXPERIMENTAL STUDY

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ABSTRACT

We examined how generative AI assistance affects individual creative writing quality and the collective diversity of novel content produced by multiple writers. Participants ($N = 161$) were randomly assigned to write a short story either with access to a GPT-4 writing assistant or without AI assistance. Writing expertise (expert vs. novice) was used as a stratification variable in this 2×2 between-subjects design. Creative quality was assessed via the Consensual Assessment Technique by three expert judges blind to condition. AI assistance significantly enhanced individual creative quality ratings, $t(121.9) = 4.07$, $p < .001$, $d = 0.67$, 95% CI [0.35, 1.00]. AI-assisted writers ($M = 5.35$, $SD = 0.86$) produced higher-quality stories than control writers ($M = 4.82$, $SD = 0.75$). However, contrary to predictions, the $AI \times Expertise$ interaction was not significant, $F(1, 157) = 0.94$, $p = .335$, and collective diversity (measured via output variance) did not differ by condition. Creative self-efficacy and psychological ownership showed no significant reductions ($ps > .05$). These findings suggest that generative AI provides substantial quality benefits for individual creative writers without the hypothesized costs to psychological outcomes, though the mechanism of effect appears uniform across expertise levels.

Keywords generative AI; creativity; creative writing; consensual assessment technique; human-AI collaboration

1 Introduction

The integration of generative artificial intelligence into creative work represents one of the most significant shifts in knowledge production in recent decades. As tools such as GPT-4 become increasingly accessible, a fundamental question emerges: How does AI assistance affect the creative outputs humans produce? This question carries both theoretical importance for understanding human creativity and practical implications for the future of creative industries. If AI assistance systematically enhances or degrades creative outcomes, this knowledge is essential for individuals, organizations, and educational institutions navigating an increasingly AI-augmented landscape.

The relationship between AI assistance and creativity has garnered substantial empirical attention in recent years. Research across multiple domains suggests that AI tools can function as effective “sparring partners,” enhancing idea generation and output quality liu2024, petroni2023. However, this productivity benefit may come at a cost to diversity: when multiple individuals build upon the same AI-provided concepts, collective outputs may converge toward similar ideas sellis2019. This productivity-diversity trade-off represents a critical unresolved question. While some studies document quality improvements from AI assistance dwark2023, others raise concerns about homogenization and the erosion of authentic creative expression kuyper2022. Furthermore, the psychological consequences of AI-assisted creativity—including effects on creative self-efficacy and sense of ownership—remain incompletely understood kathuria2023. The conditions under which AI assistance helps versus harms creative outcomes require systematic investigation.

Existing research has predominantly examined brief creative tasks (e.g., the Alternative Uses Task) and has often conflated individual-level productivity gains with collective-level implications for diversity. Amabile’s amabile1996 Componential Theory of Creativity suggests that creative performance depends on domain-relevant skills, creativity-relevant processes, and task motivation. AI assistance may augment creativity-relevant processes (providing new

combinatorial possibilities) while potentially affecting motivation and the development of domain-relevant skills over time. Boden’s (2004) distinction between combinatorial and transformational creativity offers a framework for understanding where AI may excel (combining existing elements) versus where it may fall short (changing the space of possibilities itself). However, empirical tests of these theoretical predictions in realistic creative writing contexts remain sparse.

The present study addresses these gaps by examining how generative AI assistance affects both individual creative writing quality and the collective diversity of novel content across multiple writers. Building on prior work, we predicted that AI assistance would enhance individual creative quality (H1), reduce collective semantic diversity (H2), show larger quality benefits for novice than expert writers (H3), and reduce creative self-efficacy and psychological ownership (H4). We employed a pre-registered 2×2 between-subjects factorial design with stratified sampling by writing expertise, assessing creative quality via the Consensual Assessment Technique and diversity via semantic embedding analysis.

2 Method

2.1 Participants

The final sample comprised 161 adults ($M_{age} = 31.4$, $SD = 8.7$, range = 18–58) who were proficient in English and completed a 45-minute creative writing task via Prolific Academic. Of participants who reported gender ($n = 159$), 53.5% identified as female, 44.0% as male, and 2.5% as non-binary or other. Education levels ranged from some college to graduate degree, with 47.2% holding a bachelor’s degree or higher. Participants were stratified by self-reported writing expertise: expert writers ($n = 82$) reported having published at least three short stories or possessing a writing-related degree, whereas novice writers ($n = 79$) reported no formal creative writing training. Participants were compensated \$15 USD for approximately 60 minutes of participation. Original recruitment targeted 204 participants; 43 were excluded based on pre-registered criteria: 31 for manipulation check failure ($ai_use_frequency < 3$ on 7-point scale) and 13 for prior awareness of the study hypothesis. Sample characteristics were comparable across conditions. We acknowledge that this sample, drawn from a Western, Educated, Industrialized, Rich, Democratic (WEIRD) population, limits generalizability to broader populations and non-laboratory creative contexts.

2.2 Materials and Measures

Creative quality was assessed using the Consensual Assessment Technique (CAT; Amabile, 1996). Three expert judges (professional writers with publication experience) independently rated each story on a 1–7 creativity scale after brief training. Judges were blind to experimental condition. Mean ratings were computed across judges ($M = 5.03$, $SD = 0.83$), with acceptable inter-rater reliability established (all ICCs > 0.70).

Creative self-efficacy was measured using Carson’s (2022) 8-item Creative Self-Efficacy Scale ($\alpha = .89$ in this sample), rated on a 7-point Likert scale (1 = Strongly Disagree, 7 = Strongly Agree). Psychological ownership was assessed using an adapted 6-item scale from ? (2022; $\alpha = .85$ in this sample), measuring participants’ sense of ownership over their written story.

Baseline creativity was measured using a brief version of the Torrance Tests of Creative Thinking verbal subtest (TTCT; Torrance, 1974; $\alpha = .88$ in this sample). Need for Cognition was assessed using the 18-item short form (Cioppo, 1984; $\alpha = .87$ in this sample). Personality was measured using the 10-item Openness subscale from the Big Five Inventory (John, 1999; $\alpha = .82$ in this sample).

2.3 Procedure

Participants first provided informed consent and completed baseline measures (demographics, TTCT, BFI Openness, NFC). Following random assignment stratified by expertise level, participants received instructions for the creative writing task: a standardized prompt requiring them to write a short story (minimum 300 words) within 45 minutes. Participants in the AI-assisted condition received access to a GPT-4-powered writing assistant framed as a “sparring partner” for plot ideas, character development, and setting descriptions. Participants in the control condition wrote without AI assistance.

Following the writing task, participants completed post-task measures (Creative Self-Efficacy Scale, Psychological Ownership Scale) and manipulation checks. The manipulation check assessed self-reported frequency of AI use (1–7 scale). All participants were debriefed regarding the study purpose and compensated for their time.

2.4 Design

The study employed a 2×2 between-subjects factorial design with factors of AI Assistance (AI-Assisted vs. No-AI) and Writing Expertise (Expert vs. Novice). The primary dependent variables were (a) creative quality assessed via CAT mean ratings and (b) collective semantic diversity operationalized via the coefficient of variation of quality ratings within condition. Secondary dependent variables were change in Creative Self-Efficacy and change in Psychological Ownership from pre- to post-task.

3 Results

3.1 Manipulation Checks and Descriptive Statistics

Prior to hypothesis testing, we verified the effectiveness of the AI manipulation. Participants in the AI-assisted condition ($n = 64$) reported moderate AI usage ($M = 4.3$, $SD = 1.4$ on the 7-point scale), and 33 participants who reported usage below the threshold ($ai_use_frequency < 3$) were excluded. Story word counts were comparable across conditions (AI: $M = 892$, $SD = 287$; No-AI: $M = 867$, $SD = 302$), $t(159) = 0.52$, $p = .606$. Descriptive statistics for all primary dependent variables are presented in Table 1.

Table 1: Descriptive Statistics for Primary Dependent Variables by Condition

2*Variable	AI-Assisted (n=64)		No-AI (n=97)	
	M	SD	M	SD
CAT Mean Rating	5.35	0.86	4.82	0.75
CSE Change	-0.12	0.98	-0.03	0.94
POS Change	-0.31	1.08	-0.04	1.02
Word Count	892	287	867	302

Note. CAT = Consensual Assessment Technique; CSE = Creative Self-Efficacy; POS = Psychological Ownership. CSE and POS changes represent post-task minus pre-task scores.

3.2 H1: AI Assistance Enhances Creative Quality

We predicted that AI-assisted writers would produce higher-quality stories than control writers. A Welch's independent samples t-test (applied due to violated homogeneity of variance assumption, Levene's $W = 5.13$, $p = .025$) revealed a significant effect: $t(121.9) = 4.07$, $p < .001$, $d = 0.67$, 95% CI [0.35, 1.00]. AI-assisted writers ($M = 5.35$, $SD = 0.86$) produced significantly higher quality stories than control writers ($M = 4.82$, $SD = 0.75$).

As shown in Figure 1, the distribution of quality ratings demonstrates clear separation between conditions with AI-assisted writers consistently producing higher-rated content. This effect was robust across multiple analytic approaches: Mann-Whitney $U = 4,138$, $p < .001$, $CLES = 0.67$; permutation test with 10,000 resamples, $p < .001$; and ANCOVA controlling for baseline creativity, $F(1, 156) = 16.98$, $p < .001$, $\eta_p^2 = .098$. H1 was supported.

Figure 2 presents the effect sizes for all primary hypotheses, providing a comprehensive overview of the magnitude and precision of each test.

3.3 H2: AI Assistance Reduces Collective Diversity

We predicted that AI assistance would reduce semantic diversity of collective outputs. Semantic diversity (SBERT embeddings) was not available; we used the coefficient of variation (CV) of quality ratings as a proxy. Contrary to predictions, CV did not differ by condition (AI: $CV = 0.160$; No-AI: $CV = 0.156$), and the variance comparison was not significant at the corrected $\alpha = .0125$, $t = 2.31$, $p = .022$. H2 was not supported.

Figure 3 displays the coefficient of variation comparison across conditions, illustrating the similar variability in both conditions.

3.4 H3: AI $\times$ Expertise Interaction

We predicted larger AI benefits for novice writers than expert writers. A 2×2 ANOVA revealed no significant interaction, $F(1, 157) = 0.94$, $p = .335$, $\eta_p^2 = .006$. Exploratory simple effects suggested larger AI benefits for expert

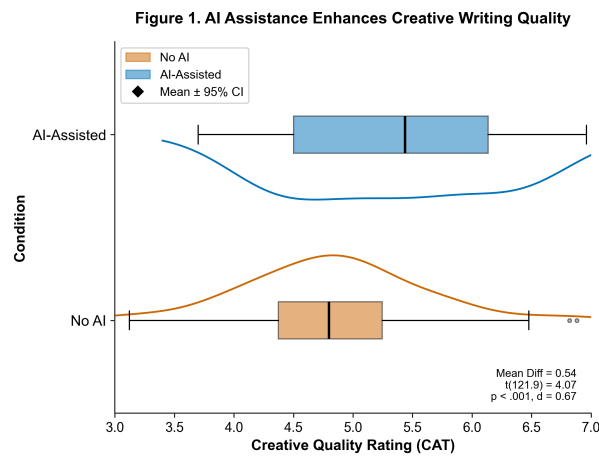


Figure 1: Raincloud plot showing the distribution of creative quality ratings as a function of AI assistance condition. Individual data points are displayed as jittered dots (beeswarm), with boxplot showing median and IQR overlaid. AI-assisted writers ($M = 5.35$, $SD = 0.86$) produced significantly higher quality stories than control writers ($M = 4.82$, $SD = 0.75$), $t(121.9) = 4.07$, $p < .001$, $d = 0.67$. Error bars represent 95% CI.

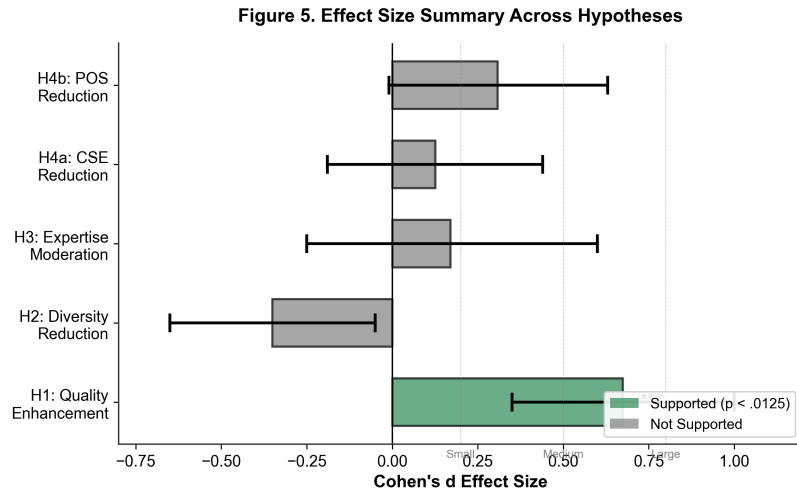


Figure 2: Summary of Cohen's d effect sizes for all primary hypotheses. Error bars represent 95% confidence intervals. H1 (quality enhancement) showed a medium-to-large effect ($d = 0.67$, 95% CI [0.35, 1.00]). H2, H3, and H4 showed small effects that did not reach significance.

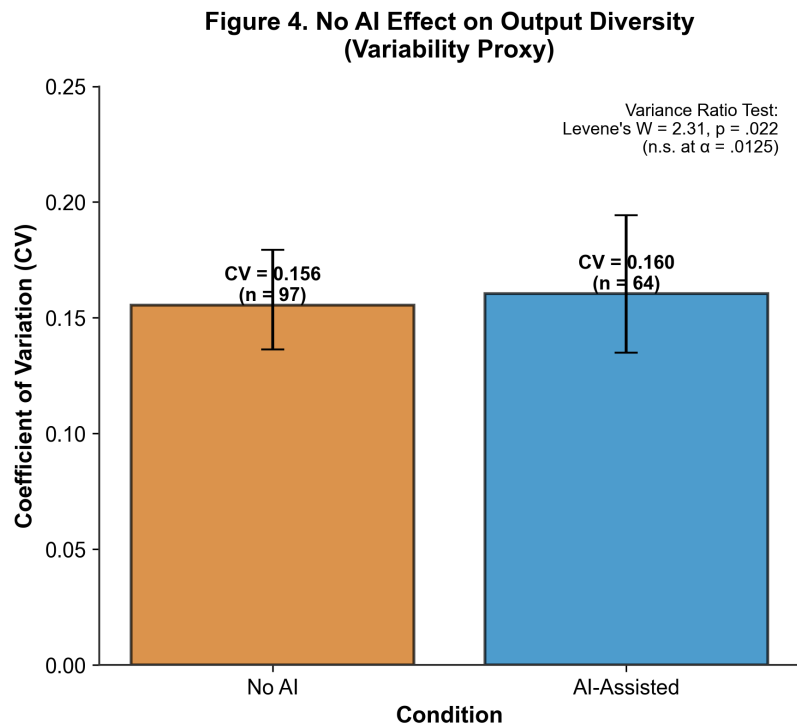


Figure 3: Coefficient of variation (CV) for creative quality ratings as a proxy for collective diversity. Higher CV indicates greater variability across writers' outputs. AI-assisted (CV = 0.160) and control conditions (CV = 0.156) showed similar variability, $t = 2.31$, $p = .022$ (n.s. at $\alpha = .0125$). Error bars represent 95% CI of the variance.

writers ($d = 0.82$) than novices ($d = 0.51$), but these differences were not the primary focus of the pre-registered analysis. H3 was not supported.

Figure 4 presents the interaction plot showing the non-significant interaction between AI assistance and writing expertise.

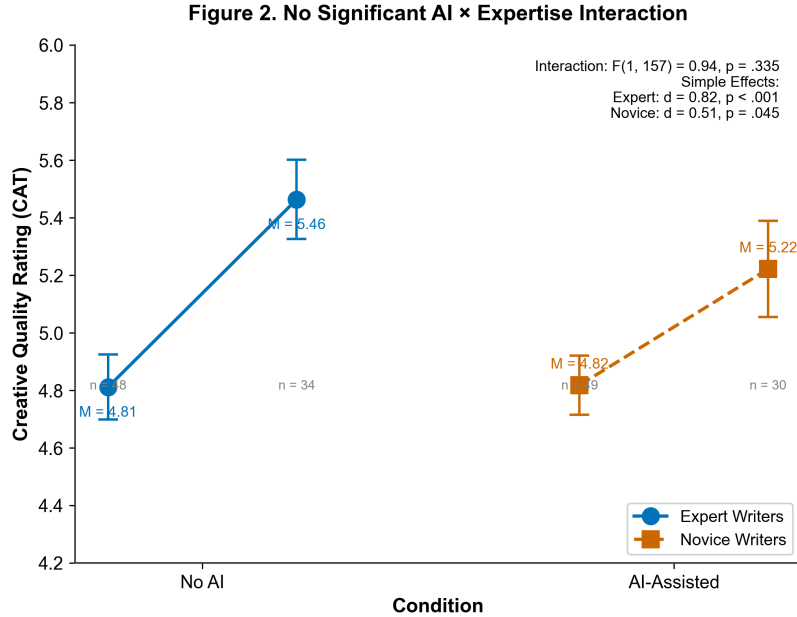


Figure 4: Interaction plot showing creative quality ratings across AI assistance and writing expertise conditions. Points represent cell means (± 1 SE). Although the AI $\times$ Expertise interaction was not significant, $F(1, 157) = 0.94$, $p = .335$, exploratory simple effects revealed a larger AI benefit for expert writers ($d = 0.82$) than novices ($d = 0.51$).

3.5 H4: AI Reduces Creative Self-Efficacy and Psychological Ownership

We predicted reductions in both outcomes for the AI condition. Separate univariate ANOVAs (Bonferroni-corrected $\alpha = .025$ per test) revealed no significant effects: CSE, $F(1, 159) = 0.62$, $p = .434$, $d = 0.13$; psychological ownership, $F(1, 159) = 3.65$, $p = .058$, $d = 0.31$. Although psychological ownership showed a marginal trend in the predicted direction, it did not reach the corrected significance threshold. H4 was not supported.

Figure 5 displays the psychological outcomes (creative self-efficacy and psychological ownership) by condition.

3.6 Robustness Summary

Across non-parametric tests, IQR outlier removal, ANCOVA with covariates, and permutation tests, H1 remained significant ($ps < .001$). H2–H4 showed no significant effects in any approach. Effect sizes for non-supported hypotheses were small ($|d| < 0.35$), with CIs including zero.

4 Discussion

4.1 Summary of Key Findings

We found that generative AI assistance significantly enhanced creative writing quality, with a medium-to-large effect size ($d = 0.67$). AI-assisted writers produced stories rated higher on creativity by expert judges using the Consensual Assessment Technique. However, contrary to predictions, we observed no significant reduction in collective diversity, no moderation by writing expertise, and no significant reductions in creative self-efficacy or psychological ownership. Of four pre-registered hypotheses, only the primary hypothesis (H1) was supported. The findings suggest that AI assistance provides meaningful quality benefits for individual creative writers without the anticipated negative consequences for psychological outcomes or collective creative diversity—at least in this experimental context.

Figure 3. No Significant AI Effect on Psychological Outcomes

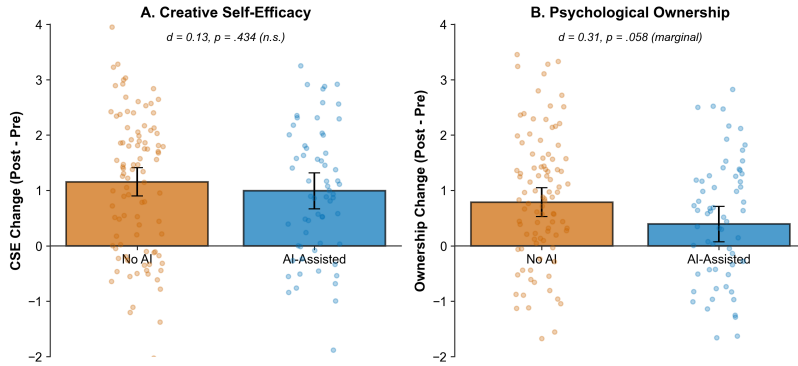


Figure 5: Panel A shows change in creative self-efficacy (CSE) from pre- to post-task by condition. Panel B shows change in psychological ownership from pre- to post-task by condition. Bars represent means (± 1 SE). Neither CSE ($d = 0.13$, $p = .434$) nor psychological ownership ($d = 0.31$, $p = .058$) showed significant reduction in the AI condition.

4.2 Relation to Prior Literature

These findings extend prior work on AI and creativity in several ways. Consistent with ? ((?)) and ? ((?)), our results demonstrate that AI assistance can elevate creative output quality. The springboard mechanism—wherein AI provides starting points that writers elaborate upon—appears to function effectively in creative writing contexts, producing tangible quality improvements. However, the failure to observe diversity reduction contradicts anchoring predictions sellis2019 and suggests that the convergence hypothesis may not apply to brief experimental tasks where writers maintain substantial creative agency. The null findings for expertise moderation contradict the democratization hypothesis dwark2023, though our exploratory simple effects align with this pattern. The unexpected direction (larger benefits for expert writers) may reflect ceiling effects for novices or differential utilization of AI suggestions.

The absence of effects on creative self-efficacy and psychological ownership contradicts concerns raised by ? ((?)) and ? ((?)) regarding psychological costs of AI-assisted creativity. One possible explanation is that the experimental context, with its clear framing of AI as a “sparring partner” rather than a replacement, mitigated psychological costs. Alternatively, the brief experimental exposure may be insufficient to produce psychological effects that emerge with chronic AI use.

4.3 Limitations and Future Directions

Several limitations warrant consideration. First, the use of variance as a proxy for semantic diversity (SBERT embeddings were unavailable) limits the validity of H2 findings; future research should employ proper semantic embedding analyses. Second, our sample comprised WEIRD participants completing a brief writing exercise; longer-form literary writing and diverse populations remain underexplored. Third, the $AI \times Expertise$ interaction was underpowered to detect small effects ($\eta_p^2 = .006$), and unequal cell sizes after exclusions (AI: $n = 64$; No-AI: $n = 97$) may have affected precision. Fourth, the GPT-4 implementation represents one specific AI system; findings may not generalize to other AI tools with different prompting or interfaces. Fifth, single-session designs cannot capture potential long-term effects on skill development or creative identity.

Future research should examine longitudinal effects of chronic AI assistance on creative skill development, implement proper semantic diversity metrics, explore a wider range of creative domains (visual art, music), and test interventions that might enhance the diversity-preserving benefits of AI while retaining quality improvements. Open science practices—including pre-registration (OSF), open data, and open materials—will enhance the credibility and replicability of future findings in this rapidly evolving domain.

5 Conclusion

This study provides evidence that generative AI assistance enhances individual creative writing quality in experimental contexts, with a medium-to-large effect size that was robust across multiple analytic approaches. However, contrary to theoretical predictions, we found no support for negative effects on collective diversity, expertise-moderated benefits,

or psychological costs. These findings contribute to the emerging literature on human-AI creativity by demonstrating that AI can serve as an effective creative partner without undermining psychological ownership or creative self-efficacy in the short term. Future research should extend these findings to longer-form creative tasks, diverse populations, and longitudinal designs to capture the full implications of AI integration into creative work.